Machine Learning

Assigment 1: Unsupervised Learning Methods

Clustering Patient Data: A Comparative Analysis

Weight	25% of final grade
Deadline	30.03.2026 (23:59 CET)
Deliverables	Jupyter Notebook + Written Report (PDF)
Dataset	Heart Disease UCI (UCI Machine Learning Repository)

Overview

In this assignment, you will apply and compare three unsupervised learning methods — **K-Means Clustering**, **Gaussian Mixture Models (GMM)**, and **Hierarchical Clustering** — to a real-world medical dataset. Your goal is to discover meaningful patient subgroups *without* using the diagnostic label, then evaluate how well your clusters align with the ground truth once it is revealed.

This mirrors a real challenge in health data science: identifying patient profiles from clinical measurements alone, before a diagnosis is available. The skills you practise here — preprocessing, dimensionality reduction, clustering, and evaluation — form the backbone of many unsupervised pipelines in healthcare.

Dataset

Heart Disease UCI Dataset

You will work with the Heart Disease dataset from the UCI Machine Learning Repository, publicly available at:

`https://archive.ics.uci.edu/dataset/45/heart+disease`

The dataset (use only processed.cleveland.data) contains 303 patient records with 13 clinical features and one target variable. You must treat the target variable as **hidden** during clustering, and only use it for evaluation afterwards.

Features (processed.cleveland.data)

Feature	Name	Description
`age`	Age	Age in years
`sex`	Sex	1 = male, 0 = female
`cp`	Chest pain type	4 types (0–3)
`trestbps`	Resting blood pressure	mm Hg on admission
`chol`	Serum cholesterol	mg/dl
`fbs`	Fasting blood sugar	$>$120 mg/dl: 1 = true, 0 = false
`restecg`	Resting ECG results	Values 0, 1, 2
`thalach`	Max heart rate achieved	Numeric
`exang`	Exercise-induced angina	1 = yes, 0 = no
`oldpeak`	ST depression	Induced by exercise vs. rest
`slope`	Slope of peak ST segment	Values 0, 1, 2
`ca`	Major vessels (0–3)	Coloured by fluoroscopy
`thal`	Thal	3 = normal, 6 = fixed defect, 7 = reversible defect
`target`	**Diagnosis (HIDDEN)**	*0 = no disease, 1–4 = disease present. Binarise as 0 vs. 1+*

Assignment Tasks

The assignment follows a single end-to-end pipeline. All team members should understand every stage, even if responsibilities are divided.

Task 1 — Data Loading & Exploration

- Load the dataset and display its shape, data types, and summary statistics.
- Identify and handle missing values. Document your strategy and justify it.
- Plot distributions of all features (histograms, box plots).
- Plot a correlation heatmap. Discuss which features appear most related.
- Keep the target column separate and do **not** use it during clustering.

Task 2 — Preprocessing & Feature Engineering

- Encode categorical features appropriately.
- Normalise or standardise features. Justify your choice of method.
- **Step A — PCA:**
  - Plot the explained variance ratio per component and cumulatively.
  - Select the number of components that explain at least 85% of variance.
  - Visualise the data in the first two principal components.

  Note: PCA is used here for visualisation purposes only. Clustering in Tasks 3–5 will be performed on the scaled feature set selected in Step B.

- **Step B — Feature selection:**
  - Split the data into training and test sets before applying feature selection. This is the only step where the target label may be used — supervised feature selection methods (as covered in the course) require it. The test set and the target label must not be used again after this point.
  - Apply at least one feature selection method covered in the course on the training set.
  - Compare the selected feature subset with the components identified by PCA. Do they capture similar structure?
  - Justify which features you will retain for clustering, and why.

Task 3 — Clustering: K-Means

- Apply K-Means for values of k from 2 to 10.
- Use the Elbow Method (inertia / cost) to identify the optimal k.
- Also compute the Silhouette Score for each k and plot it.
- Fit the final K-Means model with your chosen k and visualize the clusters in PCA space.
- Visualise the cluster assignments by projecting the data into PCA space (2D). Report and interpret the cluster centres directly in the original feature space, which features are highest or lowest for each group?

Task 4 — Clustering: Gaussian Mixture Models

- Fit GMMs with a varying number of components (2 to 10).
- Use BIC or AIC to select the optimal number of components. Plot the selected method curve.
- Visualise the cluster assignments by projecting the data into PCA space (2D). Report and interpret the cluster centres directly in the original feature space, which features are highest or lowest for each group?.
- Compare the hard cluster assignments from GMM with K-Means. Where do they agree and disagree?

Task 5 — Clustering: Hierarchical Clustering

- Compute a dendrogram using at least two linkage methods (e.g. Ward, complete, average).
- Choose a cut point on the dendrogram and justify your choice.
- Visualise the cluster assignments by projecting the data into PCA space (2D). Report and interpret the cluster centres directly in the original feature space, which features are highest or lowest for each group?.
- Discuss how hierarchical clustering differs structurally from the previous two methods.

Task 6 — Evaluation Against Ground Truth

Now reveal the target label and evaluate your clustering results:

- Compute the Adjusted Rand Index (ARI) for each method.

- Plot a confusion-style matrix showing how clusters map to the binary target (disease / no disease).
- Which method best recovered the true patient groups? Discuss why, referencing the nature of the data and the assumptions of each algorithm.

Task 7 — Written Report

In addition to your Jupyter Notebook, submit a written report of **1,500–2,500 words** addressing:

- A brief description of the dataset and preprocessing decisions.
- A comparison of the three clustering methods: their assumptions, strengths, and limitations.
- Your evaluation results and which method performed best.
- Reflection: what does the structure of the data (number of features, scale, noise) imply for the choice of clustering algorithm?
- One limitation of your analysis and how you would address it with more time.

Deliverables

Item	Description	Format
Jupyter Notebook	Fully executed notebook with all outputs visible. Code must be clean and commented.	`.ipynb`
Written Report	1,500–2,500 words. Figures may be embedded. Reference list not included in word count.	`.pdf`

Submit both files as a single `.zip` archive named: `TeamXX_UnsupervisedLearning.zip`

Grading Rubric

Total: 100 points (counts as 25% of the final grade)

Section	Pts	Criteria
Task 1: Data Exploration	**10**	
Loading & shape check	2	Data loaded correctly; shape, dtypes, and nulls reported.
Distributions	4	Histograms/box plots for all features. Observations discussed in markdown.
Correlation analysis	4	Heatmap present. Notable correlations identified and discussed.
Task 2: Preprocessing	**15**	
Encoding & scaling	4	Correct encoding. Scaling method justified.
PCA application	6	Variance plot correct. Component count justified. 2D scatter shown.

Section	Pts	Criteria
Feature selection	5	At least one method applied. Comparison with PCA discussed. Final representation justified.
Task 3: K-Means	**15**	
Elbow & silhouette plots	6	Both plots produced and interpreted. Optimal k selected with justification.
Final model & visualisation	5	Cluster assignments plotted in PCA space. Centres reported.
Cluster characterisation	4	Feature-level description of each cluster provided.
Task 4: GMM	**15**	
BIC/AIC model selection	6	Both curves plotted. Optimal components selected and justified.
Visualization	5	Cluster assignments plotted in PCA space. Centres reported.
Comparison with K-Means	4	Agreements and disagreements discussed with examples.
Task 5: Hierarchical Clustering	**15**	
Dendrogram	6	Two linkage methods compared. Dendrogram plotted clearly.
Cut & cluster visualization	5	Cut justified. Clusters visualised in PCA space.
Structural comparison	4	Conceptual differences vs K-Means/GMM discussed.
Task 6: Evaluation	**15**	
ARI scores	9	Computed correctly for all three methods. Results tabulated.
Confusion matrix & discussion	6	Matrix plotted. Best method identified and explanation is data-driven.
Task 7: Written Report	**15**	
Methodology & preprocessing	4	Decisions explained clearly with justification.
Algorithm comparison	5	Assumptions and trade-offs of all three methods discussed accurately.
Evaluation & reflection	4	Results interpreted; limitation identified; improvement proposed.
Clarity & presentation	2	Well-structured, readable, free of major errors.
TOTAL	**100**	

Notes & Academic Integrity

- All team members are expected to understand and be able to explain every part of the submission.
- You may use `scikit-learn`, `scipy`, and standard Python data science libraries. List all

libraries used.

- AI coding tools may be used to assist with syntax, but the analytical decisions and written report must be your own. Indicate clearly if AI tools were used.

Advice for Teams

Divide the pipeline by task (one member per task), but hold a joint review session before submission. Tasks 6 and 7 should be a shared team effort, since they require synthesising results from all previous tasks. A notebook that runs end-to-end without errors is a minimum requirement for full marks.